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Modeling with Ease: End-to-End Learning in Model Studio

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#SASinnovate

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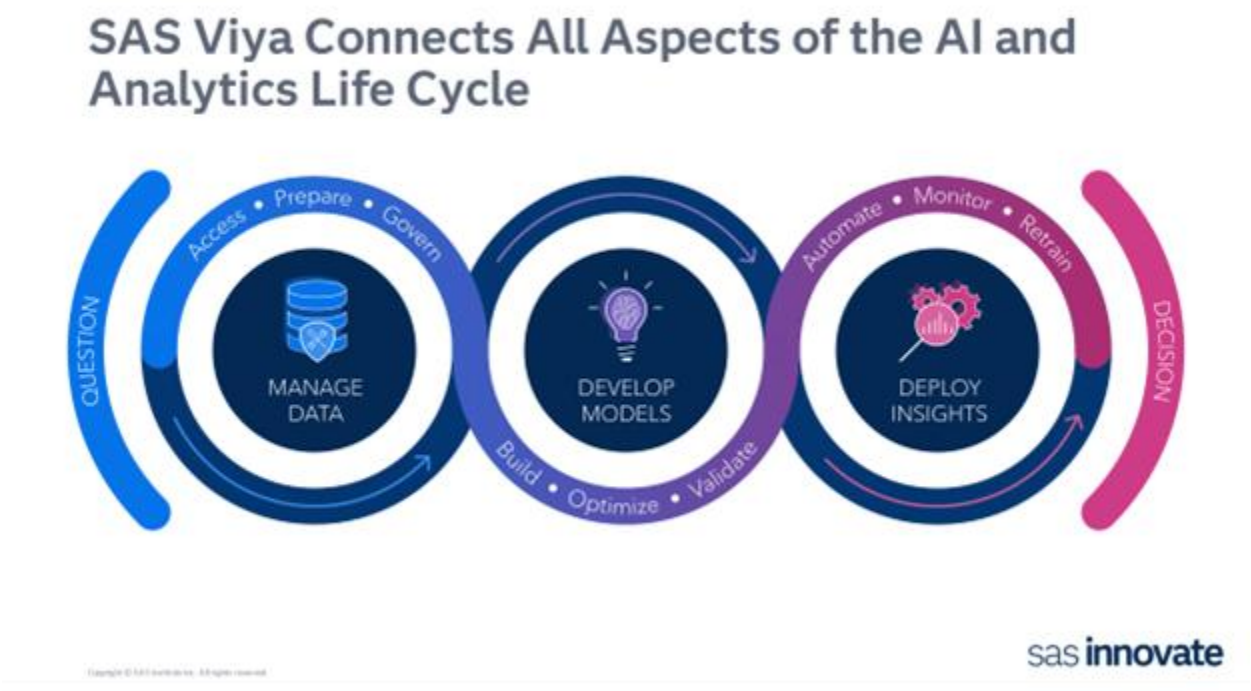
Introduction

The three phases of the analytics life cycle are ***data***, ***discovery***, and ***deployment***. Recognizing and fully supporting all three is necessary to generate impactful insights that come from transforming data into value.



SAS Drive

SAS Drive enables you to quickly access the items and applications that you work with in SAS Viya. The availability of the features on the page depends on your SAS license and the permissions that have been assigned to you by your administrator.

SAS Drive uses the standard sign-in window for SAS applications. To display the sign-in window, enter the URL that is provided by your instructor. After you sign in, you see SAS Drive.

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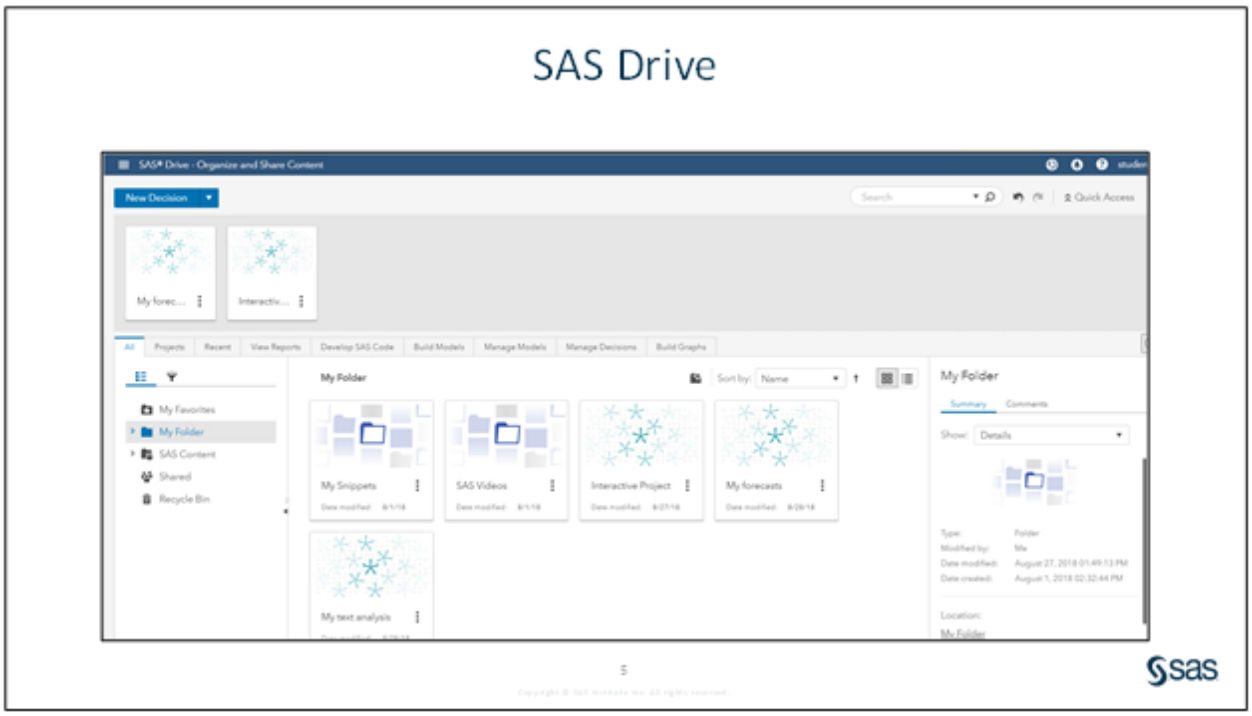

- The application bar at the top enables you to do things such as search for items, view your recent items, and access Help.
- The menu tabs in the main content area are shortcuts that open SAS applications. Note that the tabs represent functional content areas in the analytic workflow. We spend most of our time in this presentation on the Build Models and Manage Models tabs.
- The directory on the left enables you to organize and share your analyses.

For more information about SAS Drive, you can access the SAS Help Center of SAS Drive using the question mark (Help menu icon) on the right side of the application bar at the top of the screen.

Model Studio

Model Studio, included in SAS Viya, is an integrated visual environment that provides a suite of analytic tools to facilitate end-to-end data mining, text, and forecasting analyses. The tools supported in Model Studio are designed to take advantage of the SAS Viya programming and cloud processing environments to deliver and distribute the results of analyses, such as champion models, score code, and results.

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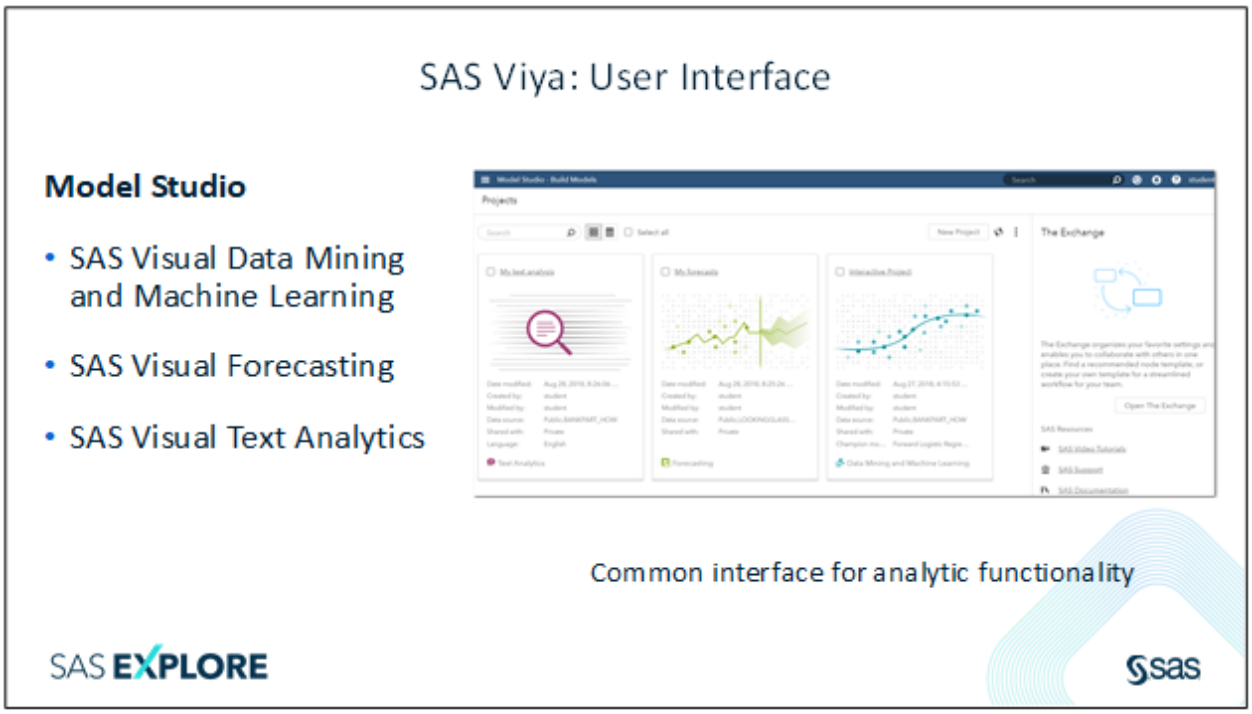

Model Studio is a common interface that contains the following SAS solutions:

- SAS Visual Forecasting
- SAS Visual Data Mining and Machine Learning in Model Studio
- SAS Visual Text Analytics

The availability of the functionality in Model Studio depends on your SAS license and the permissions that have been assigned to you by your administrator. You can access the SAS Help Center/Model Studio for more information using the question mark (Help menu icon) on the right side of the application bar.

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Preliminary Information

VS_BANK Data Dictionary

The data to be used for analysis in this workshop consists of observations taken on a large financial services firm's accounts. Accounts in the data represent consumers of home equity lines of credit, automobile loans, and other types of short- to medium-term credit instruments.

The data have been anonymized and transformed to conform to the following description:
A campaign interval for the bank runs for half a year. A campaign represents all marketing efforts that provide information about and motivate the contracting (purchase) of the bank's financial services products. Campaign promotions are categorized into direct and indirect. *Direct promotions* consist of sales offers to a particular account that involve an incentive. *Indirect promotions* are marketing efforts that do not involve an incentive.

In addition to the account identifier (name: **account**, label: **Account ID**), the data include the following:

A ***target variable*** quantifies account responses over the current campaign season.

Name	Label	Description
B_TGT	tgt Binary New Product	A binary target variable. Accounts coded with a 1 contracted for at least one product in the previous campaign season. Accounts coded with a zero did not contract for a product in the previous campaign season.
CNT_TARGET	tgt Count Number New Products	A count (interval) target variable. This is the number of new products purchased by a banking customer. If this number is nonzero then B_TGT is 1. We will note use this target in this current analysis.

Categorical valued inputs summarize account-level attributes related to the propensity to buy products and other characteristics related to profitability and creditworthiness. These variables have been transformed to anonymize account-level information and to mitigate quality issues related to excessive cardinality.

Name	Label	Description
CAT_INPUT1	Category 1 Account Activity Level	A three-level categorical variable that codes the activity of each account. • *X* à high activity. The account enters the current campaign period with a lot of products. • *Y* à average activity. • *Z* à low activity.
CAT_INPUT2	Category 2 Customer Value Level	A five-level (*A* through *E*) categorical variable that codes customer value. For example, the most profitable and creditworthy customers are coded with *A*.

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ACCOUNT	Account ID	This is a unique ID for each bank customer account. This will not be used in the analysis as an input variable

Interval valued inputs provide continuous measures on account-level attributes related to the recency, frequency, and sales amounts (RFM). These variables have been transformed to anonymize account-level information. All measures below correspond to activity prior to the current campaign season.

Name	Label	Description
RFM1	RFM1 Average Sales Past 3 Years	Average sales amount attributed to each account over the past three years
RFM2	RFM2 Average Sales Lifetime	Average sales amount attributed to each account over the account's tenure
RFM3	RFM3 Avg Sales Past 3 Years Dir Promo Resp	Average sales amount attributed to each account in the past three years in response to a direct promotion
RFM4	RFM4 Last Product Purchase Amount	Amount of the last product purchased
RFM5	RFM5 Count Purchased Past 3 Years	Number of products purchased in the past three years
RFM6	RFM6 Count Purchased Lifetime	Total number of products purchased in each account's tenure
RFM7	RFM7 Count Prchsd Past 3 Years Dir Promo Resp	Number of products purchased in the previous three years in response to a direct promotion
RFM8	RFM8 Count Prchsd Lifetime Dir Promo Resp	Total number of products purchased in the account's tenure in response to a direct promotion
RFM9	RFM9 Months Since Last Purchase	Months since the last product purchase
RFM10	RFM10 Count Total Promos Past Year	Number of total promotions received by each account in the past year
RFM11	RFM11 Count Direct Promos Past Year	Number of direct promotions received by each account in the past year
RFM12	RFM12 Customer Tenure	Customer tenure in months

Demographic variables describe the profile of each account in terms of income, homeownership, and other characteristics.

Name	Label	Description

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DEMOG_AGE	Demog Customer Age	Average age in each account's demographic region
DEMOG_GENF	Demog Female Binary	A categorical variable that is 1 if the primary holder of the account is female, and 0 otherwise
DEMOG_GENM	Demog Male Binary	A categorical variable that is 1 if the primary holder of the account is male, and 0 otherwise
DEMOG_HO	Demog Homeowner Binary	A categorical variable that is 1 if the primary holder of the account is a homeowner, and 0 otherwise
DEMOG_HOMEVAL	Demog Home Value	Average home value in each account's demographic region
DEMOG_INC	Demog Income	Average income in each account's demographic region
DEMOG_PR	Demog Percentage Retired	The percentage of retired people in each account's demographic region

Preprocessed version of inputs are also included in this dataset. These are new variables derived from the original variables and added to the dataset by another analyst. It can be useful to perform transformations to modify variables and prepare them for predictive modeling. This can be done easily in Model Studio, but sometimes the data already has these variables. In this workshop we will keep it simple and skip using the preprocessed variables.

Name	Label	Description
I_RFM#	i_rfm#	These are imputed versions of the RFM variables. They might been imputed using the mean or the median, but it's not obvious from the data so we would have to ask whoever did the original analysis.
LOGI_RFM#	logi_rfm#	These are log-transformed versions of the imputed RFM variables above.
R_DEMOG_...	demog_r …	These are recoded/replaced versions of the demographic variables. It's not obvious from the data how the recoding/replacement was performed.
RI_DEMOG_...	demog_ri	These are recoded/replaced and imputed versions of the demographic variables. It's not obvious from the data how the recoding/replacement was performed. The imputation was probably done based on the mode but it's not clear from the data.

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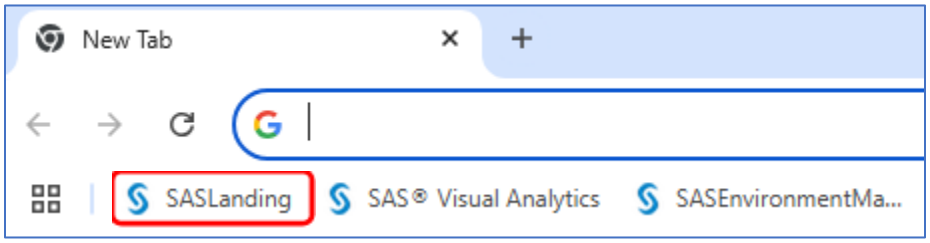
Hands-On Workshop Sign-In

Creating a Project and Building, Fitting, and Comparing Predictive Models Using Pipelines in Model Studio

This demonstration illustrates loading data into SAS Viya and building predictive modeling pipelines using Model Studio. The demonstration continues with model comparison in Model Studio and finishes with model deployment in SAS Model Manager.

1. Follow the instructions given by your instructor to access the image desktop.
2. Select the **Google Chrome** shortcut.
3. Select **SASLanding**.



4. Sign in to SAS using the user name **student** and the password **Metadata0** (these credentials should be automatically filled in).
6. Select **No** when asked about assumable groups. We don't need to be an administrator to perform this analysis.

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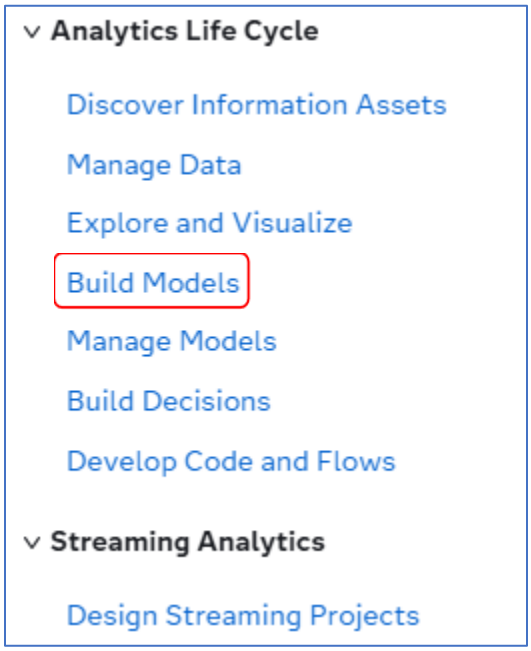
Create a Model Studio Project

1. Click the **applications menu** icon in the top left and select **Build Models**. This opens the Model Studio web application.



2. From the Model Studio Projects page, select **New Project** (in the upper right corner) to create a new modeling project.
3. Name your project **Model Studio Workshop**.
4. Under the Data option, click **Browse** to choose a data set for this modeling project.
5. From the Choose Data window find the **VS_BANK** dataset in the **Public** library and select **OK** in the bottom right.

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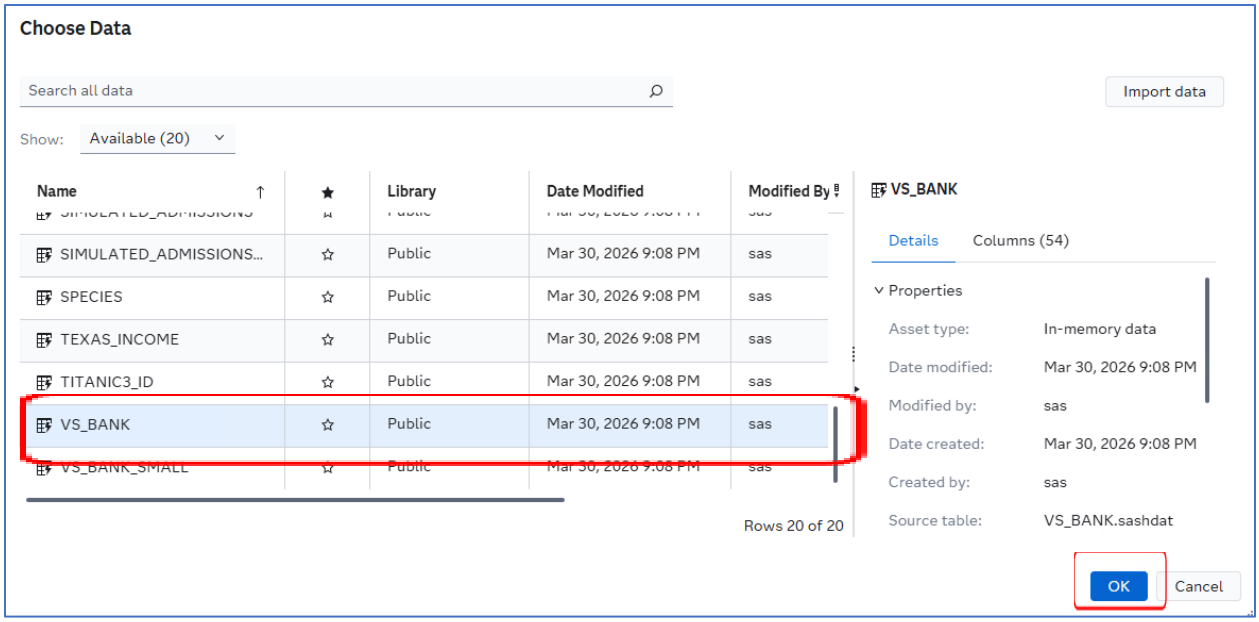
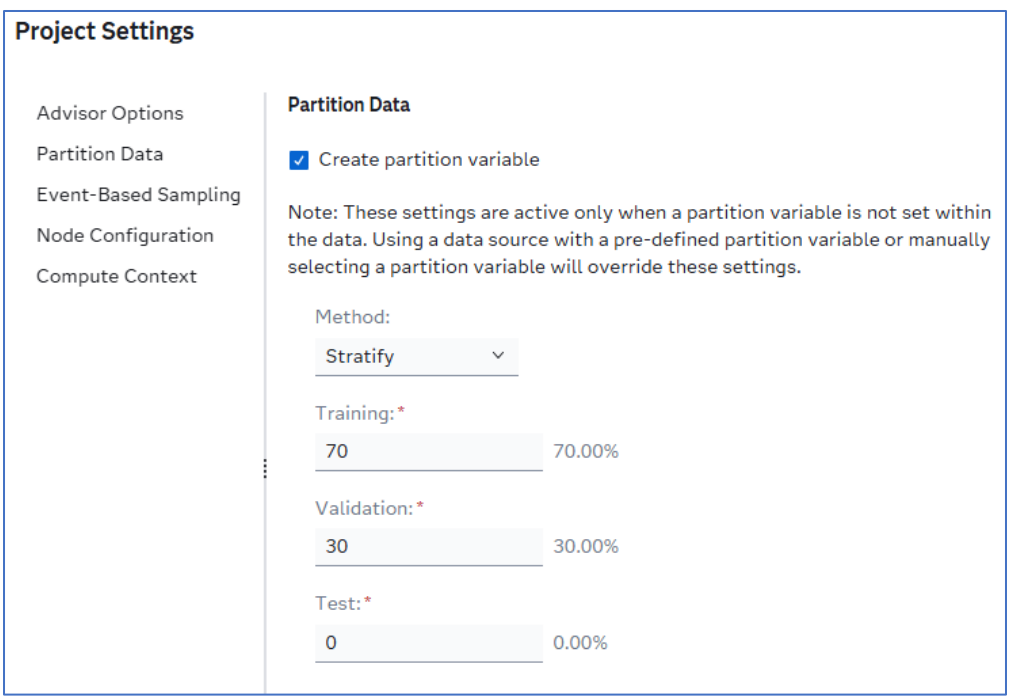


In this workshop we have already loaded the data into memory for your convenience. If you want to load the data into memory yourself, there are directions at the end of this document in **the Background Information** section.

6. Click **Advanced** to open additional New Project settings.
7. Select **Partition Data** to modify partition options.
8. Notice that a partition is automatically created. Change the Training partition to **70%**, the Validation partition to **30%**, and the Test partition to **0%**.



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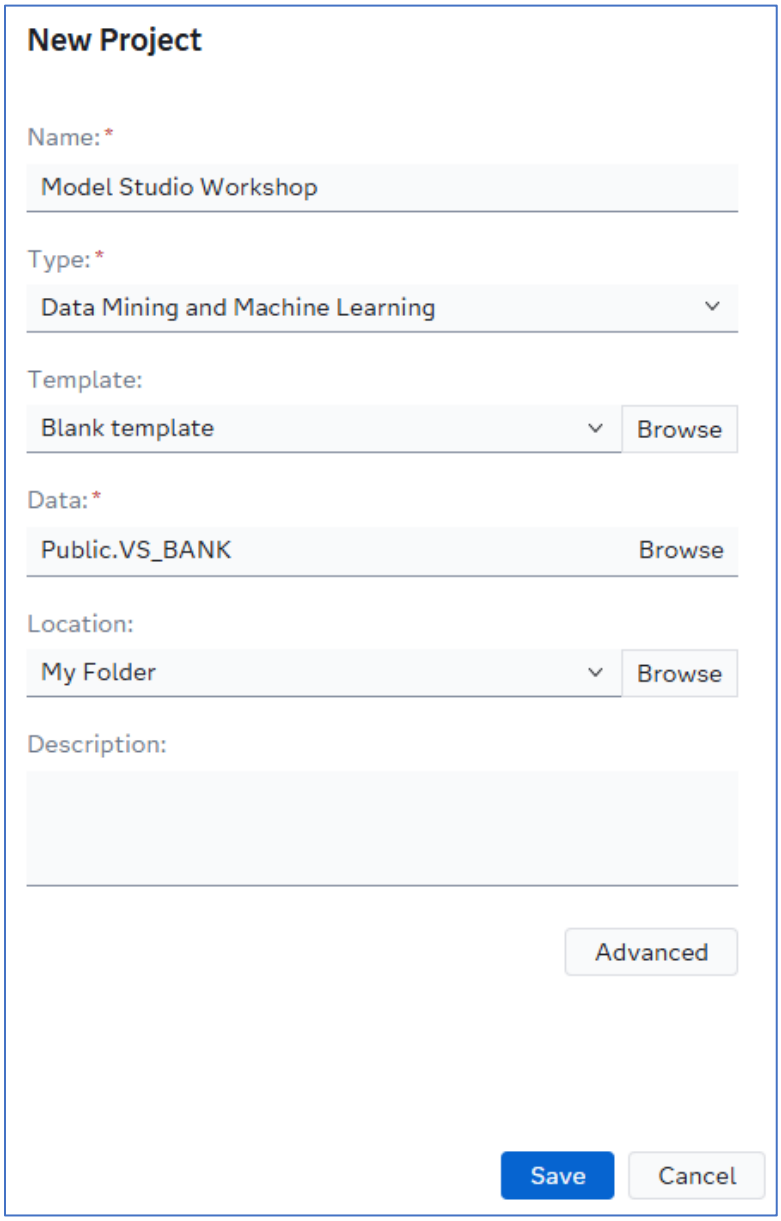
13. Click **Save** to return to the New Project window.
14. Click **Save** in the New Project window to create the modeling project. Your settings should match the screenshot below.



15. In the Data pane, click the button in the top-left of the table to select all variables, and then in the **Multiple Variables** pane on the right change the **Role** from Mixed Values to **Rejected**. The screenshot below shows what this should look like after you make the changes.

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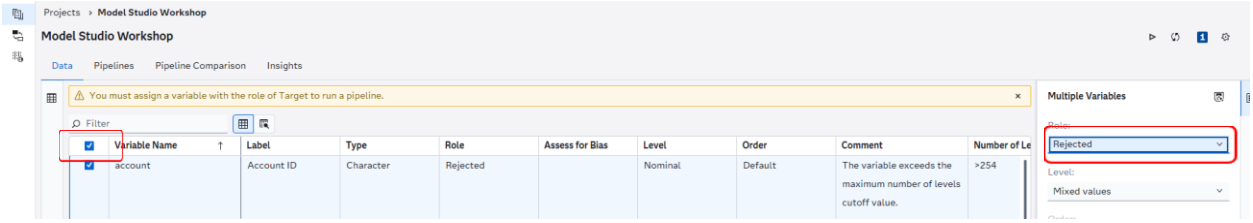

16. Now that we have rejected all of the unwanted variables, let's select our input and our target. First click the button in the top left of the table a second time to unselect all of the variables.

17. Now select only **b_tgt** (**Binary New Product**) and set its role to **Target**. It is important that we only have one target variable, and it's also important that we reject the other possible target, (cnt_tgt). Unselect **b_tgt** after making this change.

18. Now select the input variables by clicking on the check-box to the left of the variable name and then change the **Role** from **Rejected** to **Input** in the **Multiple Variables** pane on the right. We want all the original (unmodified) variables. The easiest way to select them is to hold shift while clicking on the boxes to select a group of them at a time (they are sorted in alphabetical order). We have three groups to bulk select, the **cat_input** group, the **demog_** group, and the **rfm** group. You can also individually select each variable on the list below:

    - **cat_input1**
    - **cat_input2**
    - **demog_age**
    - **demog_genf**
    - **demog_genm**
    - **demog_ho**
    - **demog_homeval**
    - **demog_inc**
    - **demog_pr**
    - **rfm1**
    - **rfm2**
    - **rfm3**
    - **rfm4**
    - **rfm5**
    - **rfm6**
    - **rfm7**
    - **rfm8**
    - **rfm9**
    - **rfm10**
    - **rfm11**
    - **rfm12**

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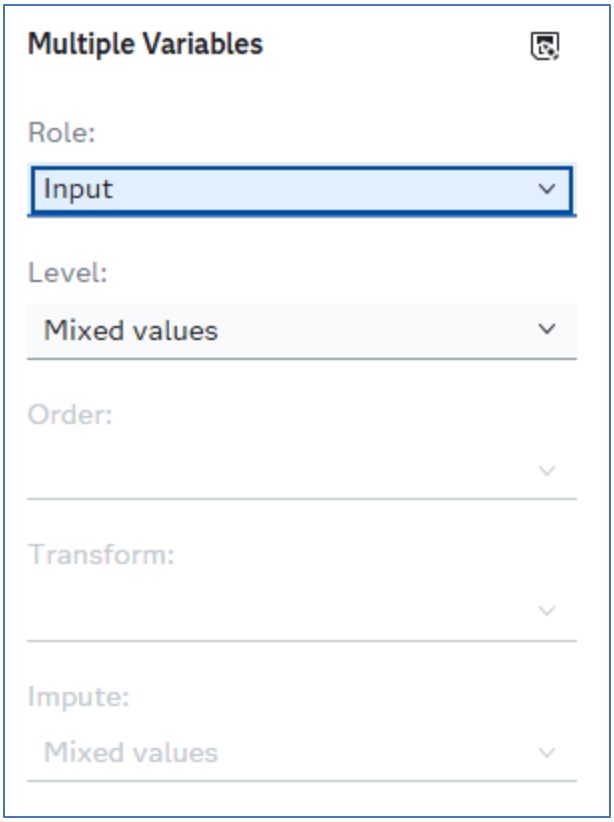

Note: If you click on the check-box to the left of a variable you can select multiple variables, but if you click on the name of the variable in the table it will deselect all other variables and select only the name you clicked on.

Now that we have setup the metadata for this project (the target variable and the input variables) we are ready to move on and start building modeling pipelines.

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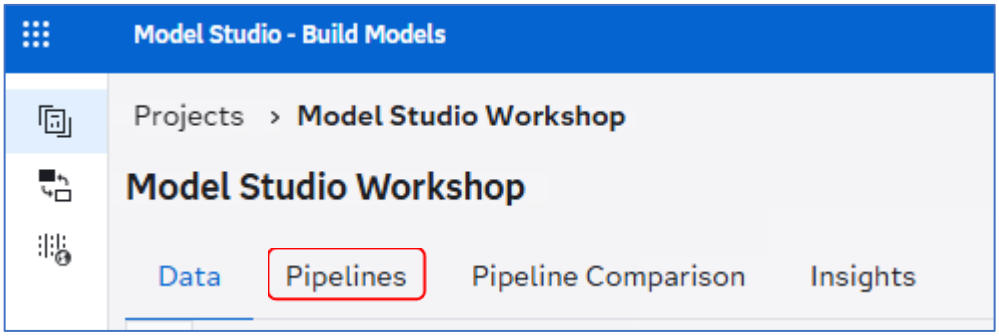
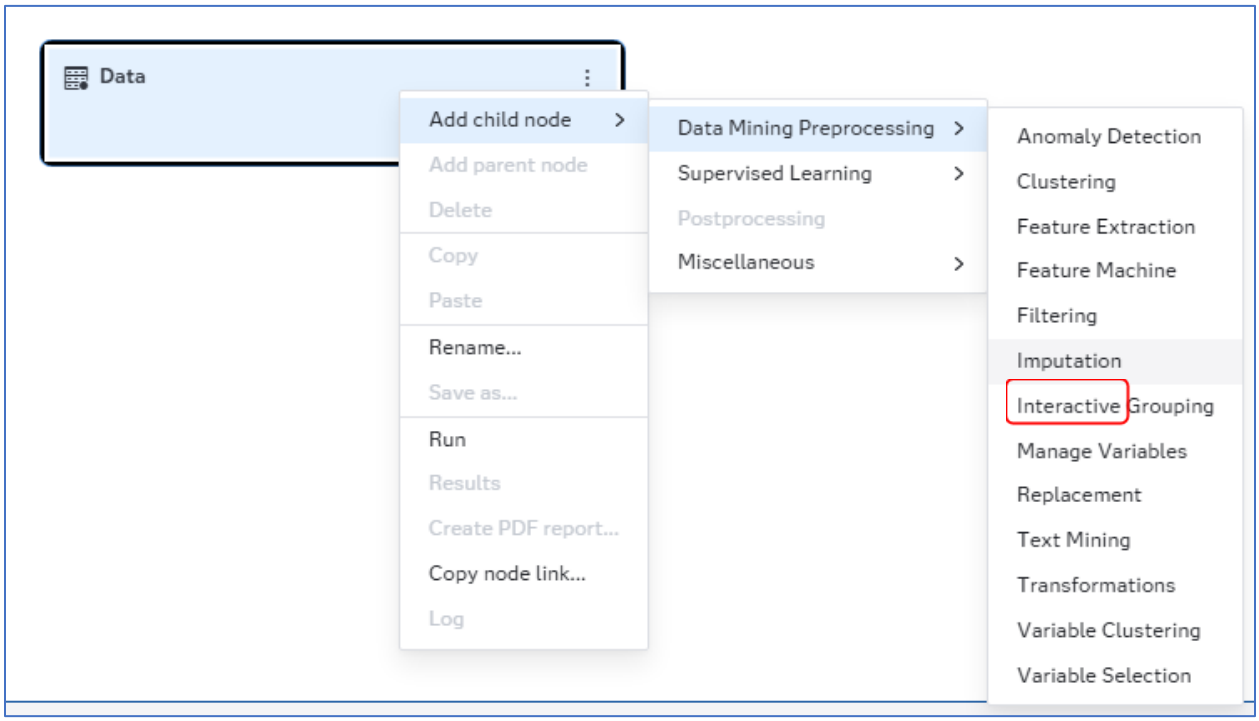
Build and Fit Predictive Models Using Pipelines with Model Interpretability Plots

1. Select **Pipelines** on the top left of the page.



2. The pipeline starts out with just a Data node. Right-click the **Data** node and select **Add child node** > **Data Mining Preprocessing** > **Imputation**.



3. Click the **Imputation** node to display the settings for imputing missing values. By default, missing class inputs are replaced with the mode (**Count** refers to the one with the highest count), and missing interval inputs are replaced with the mean.
4. Right-click the **Imputation** node and select **Add child node** > **Data Mining Preprocessing** > **Variable Selection**.
5. Click the **Variable Selection** node to display the settings. From the **Combination criterion** menu in the right pane, select **Selected by a majority**. Leave **Fast Supervised Selection** on and turn on **Linear Regression**

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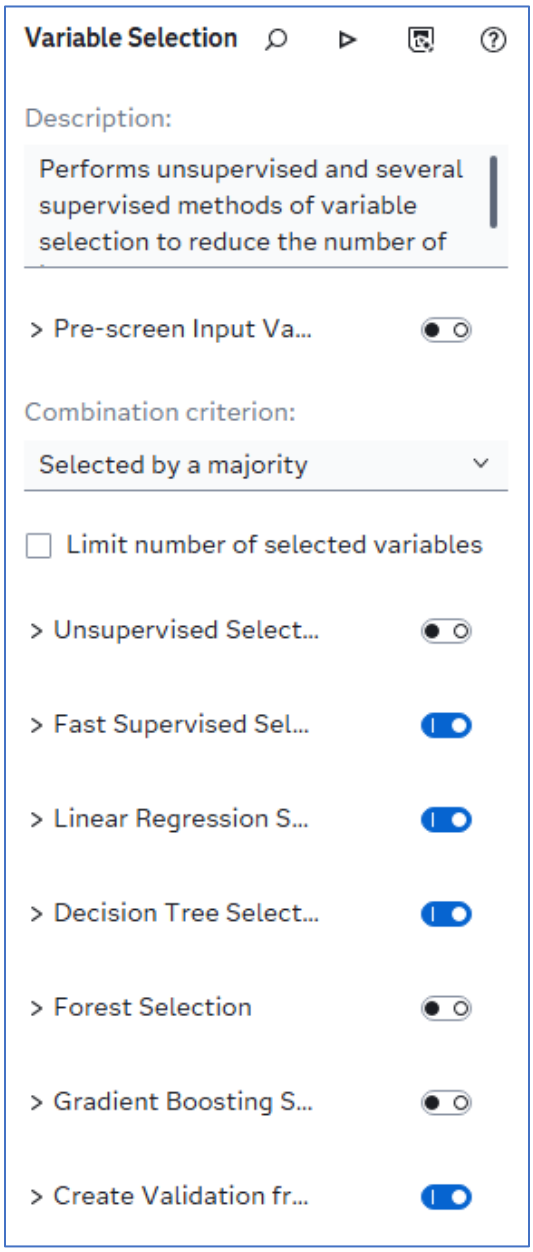
Selection and **Decision Tree Selection** by clicking the toggle switch next to those methods.



6. Right-click the **Variable Selection** node and select **Add child node** > **Supervised Learning** > **Neural Network**.
7. Click the **Neural Network** node and under **Hidden Layer Options** set **Number of neurons per hidden layer** to **1**.

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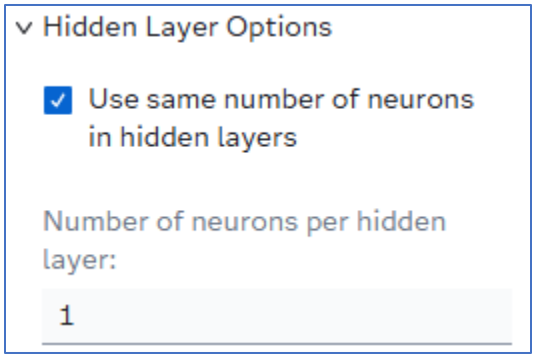
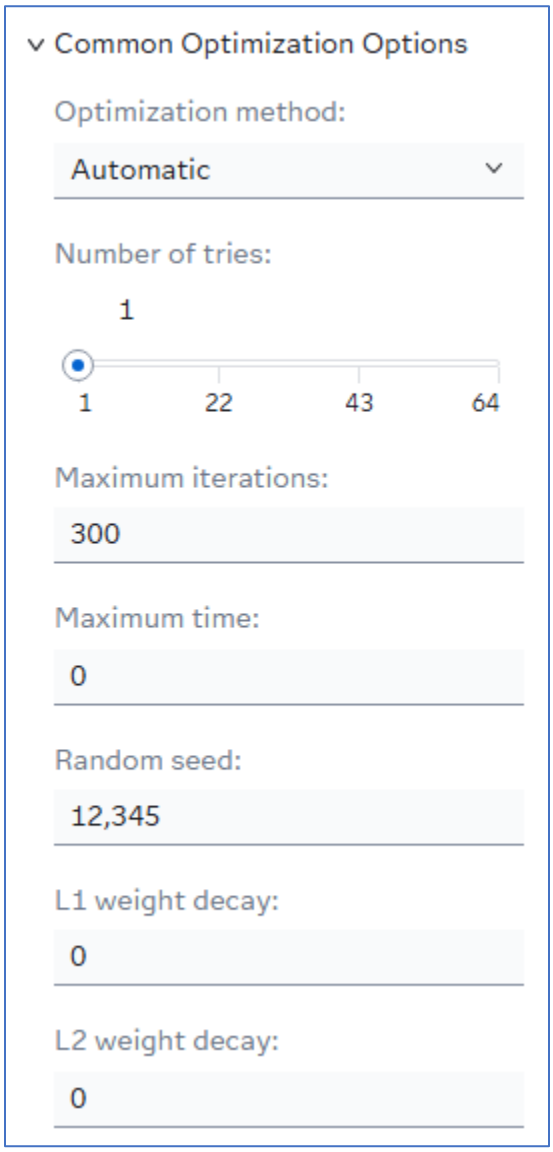


8. Open **Common Optimization Options** and set **L2 weight decay** to **0**.



9. Scroll down to the bottom of the options on the right. Under **Post-training Properties**, open **Model Interpretability**. Below that, open **Global Interpretability** and **Local Interpretability**.

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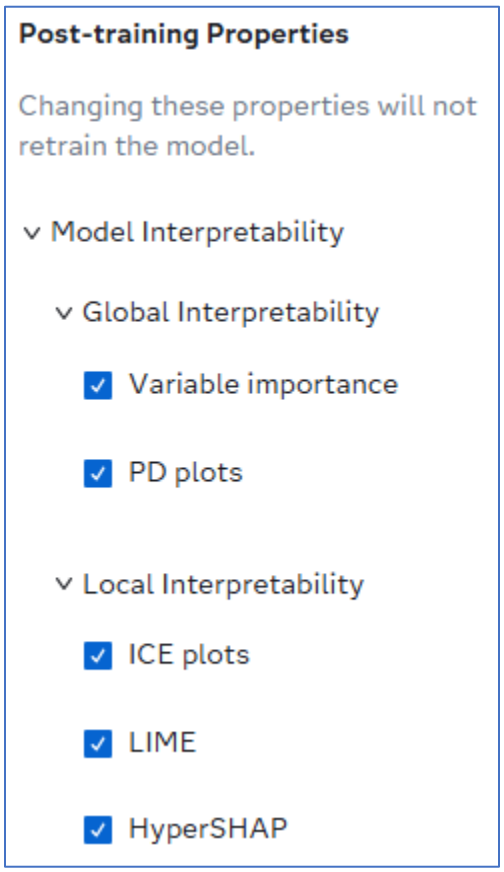
10. Select **Variable importance** and **PD plots** under **Global Interpretability**. This turns on a Variable Importance table as well as a Partial Dependence plot.
11. Select **ICE plots**, **LIME**, and **HyperSHAP** under **Local Interpretability**. This turns on the Individual Conditional Expectation plots, the Local Model-Agnostic Explanations, and the Shapley Values plots.



12. Open the **Nodes** menu on the left of the pane and expand the **Supervised Learning** list.

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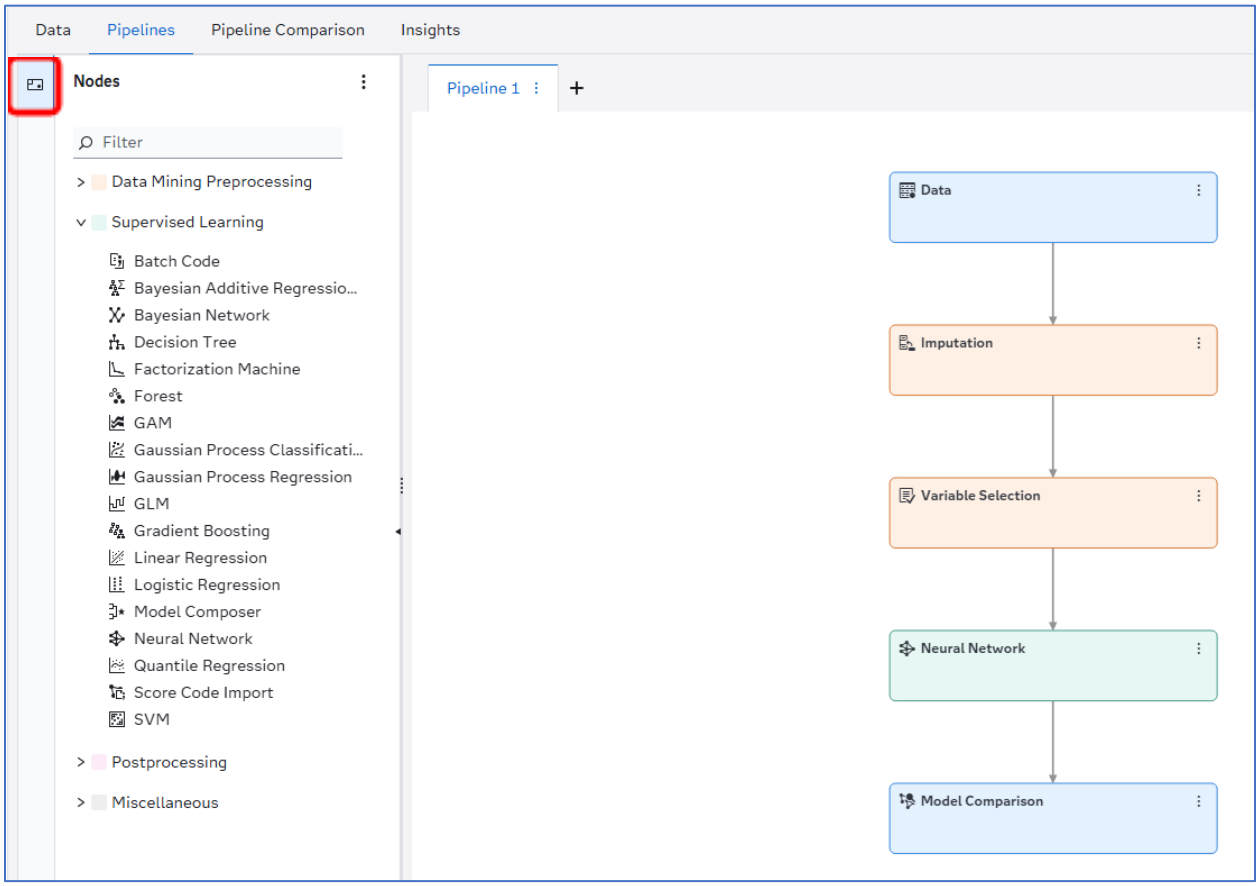
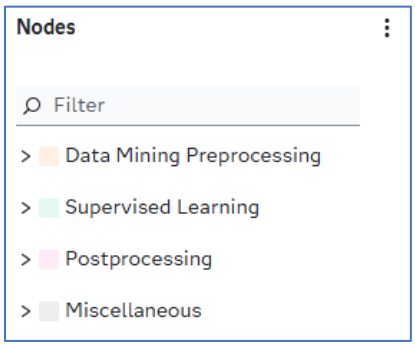


The *Data Mining Preprocessing* nodes provide tools for creating features, selecting variables, and changing values in the data. This includes some tools for unsupervised learning, although the graphical interface is fundamentally designed for supervised learning.

The *Supervised Learning* nodes provide tools for building supervised machine learning models.

The *Postprocessing* node enables you to create an ensemble of supervised machine learning models.

The *Miscellaneous* nodes enable you to explore data, incorporate SAS or open-source code into the pipeline, and integrate the pipeline with other SAS tools. We explore the Open Source Code node in this demonstration.



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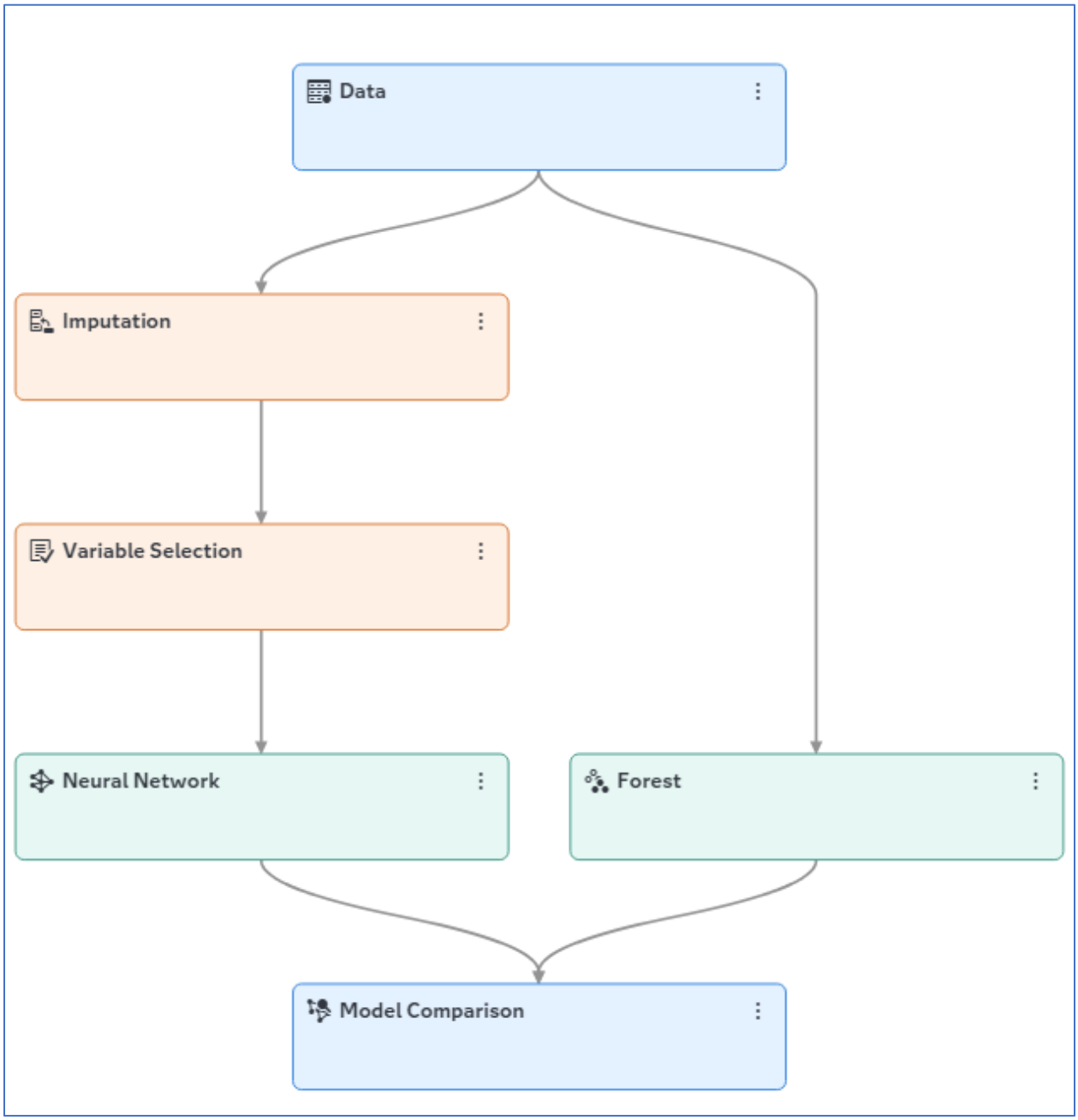
Although we won't have time to discuss most of the nodes available in Model Studio, you are encouraged to experiment by adding any of the nodes that interest you to the pipeline.

13. Drag and drop the **Forest** node from the Supervised Learning list onto the **Data** node. (You can hide the Nodes menu to gain space for the pipelines.)

Note: The neural network model requires imputation because it ignores any rows with missing values. The forest model can handle missing values and has built-in variable selection, so it can skip the Imputation and Variable Selection nodes.



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14. Click **Run Pipeline** in the top right.
15. When the Variable Selection node finishes running, right-click the **Variable Selection** node and select **Results**.

The Variable Selection table indicates which variables are kept as inputs and which variables are rejected.

Variable Selection

Name	Variable Label	Variable Level	Role	Reason
B_TGT	tgt Binary New Product	BINARY	TARGET	
DEMOG_HOMEVAL	demog Home Value	INTERVAL	INPUT	
RFM11	rfm11 Count Direct Promos Past Year	INTERVAL	INPUT	
RFM2	rfm2 Average Sales Lifetime	INTERVAL	INPUT	
RFM5	rfm5 Count Purchased Past 3 Years	NOMINAL	INPUT	

The Variable Selection Combination Summary table indicates which methods chose to keep or reject the input variables. In this case, the Fast Supervised selection, the Linear Regression selection, and the Decision Tree selection methods were active. If a majority of the methods fail to select an input, that input is set to **REJECTED** in the Variable Selection table.

Variable Selection Combination Summary

Name	Variable ...	Fast	Linear R...	Decision ...	Input	Rejected	Output R...
CAT_INPUT 2	category 2 Customer Value Level	REJECTED	INPUT	REJECTED	1	2	REJECTED
DEMOG_GENF	demog Female Binary	REJECTED	REJECTED	REJECTED	0	3	REJECTED
DEMOG_GENM	demog Male Binary	REJECTED	REJECTED	REJECTED	0	3	REJECTED
DEMOG_HO	demog	REJECTED	REJECTED	REJECTED	0	3	REJECTED

16. Close the results of the Variable Selection node.
17. Right-click the **Neural Network** node and select **Results**.

The Network Diagram plot provides a visual of the neural network architecture but does not provide any way to interpret the model. We will explore model interpretability plots for the neural network results shortly.

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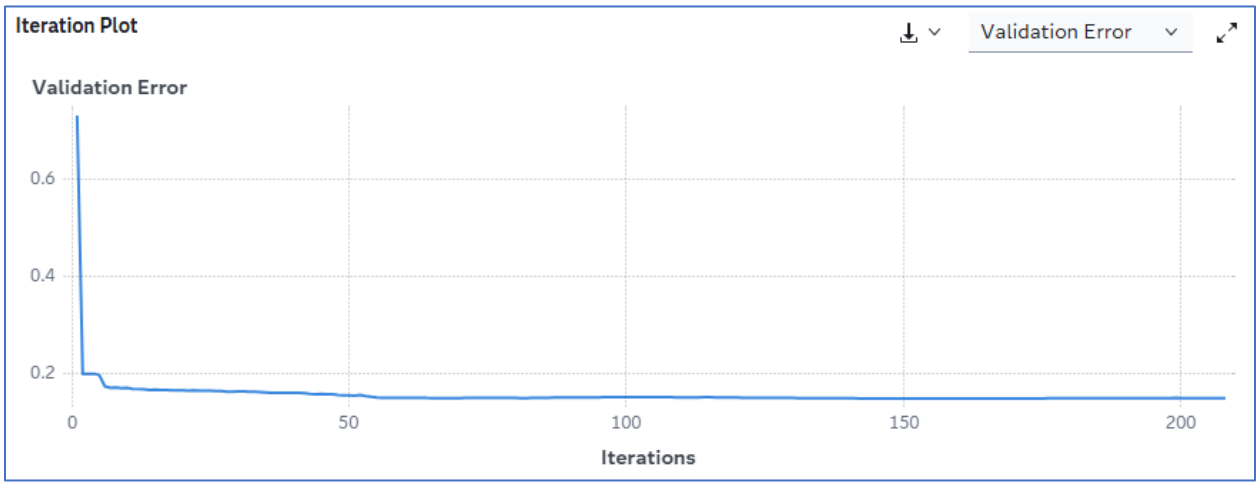
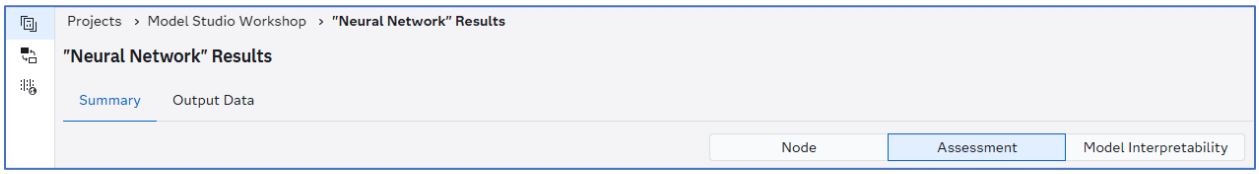
The Iteration plot indicates that the model improves performance on validation data (new data that it has not seen before) as the neural network is trained.



18. Select **Assessment** in the top center to look at model assessment plots and tables that can be created for all predictive models in Model Studio.

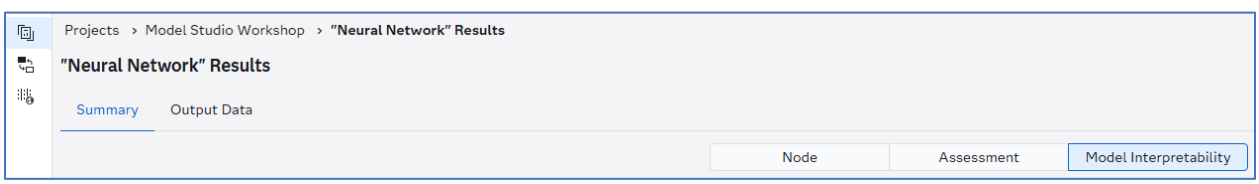


19. Click the **Expand** button on the Fit Statistics table to see a collection of assessment statistics for the model.

Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	Divisor ...	Root Av...	Misclas...	Multi-Cl...	KS (You...	Area Un...	Gini Coe...	Gamma	Tau	KS Cutoff	KS at Us...	Mis
b_tgt	TRAIN	1	1	742,026	0.1067	742,026	0.3267	0.1482	0.3461	0.5791	0.8631	0.7261	0.7339	0.2320	0.2100	0.4058	0
b_tgt	VALIDATE	0	0	318,012	0.1066	318,012	0.3266	0.1484	0.3458	0.5813	0.8634	0.7268	0.7345	0.2322	0.2100	0.4058	0

20. Close the Fit Statistics table and click the tab for **Model Interpretability**. This tab appears because we are using some of the model interpretability tools in Model Studio.



21. Explore the different types of model interpretability plots available in SAS Viya. For example, here is the Partial Dependence plot for the imputed **rfm5** variable. (For an automated description of each of the plots, click the **info** button circled below for the corresponding plot.)

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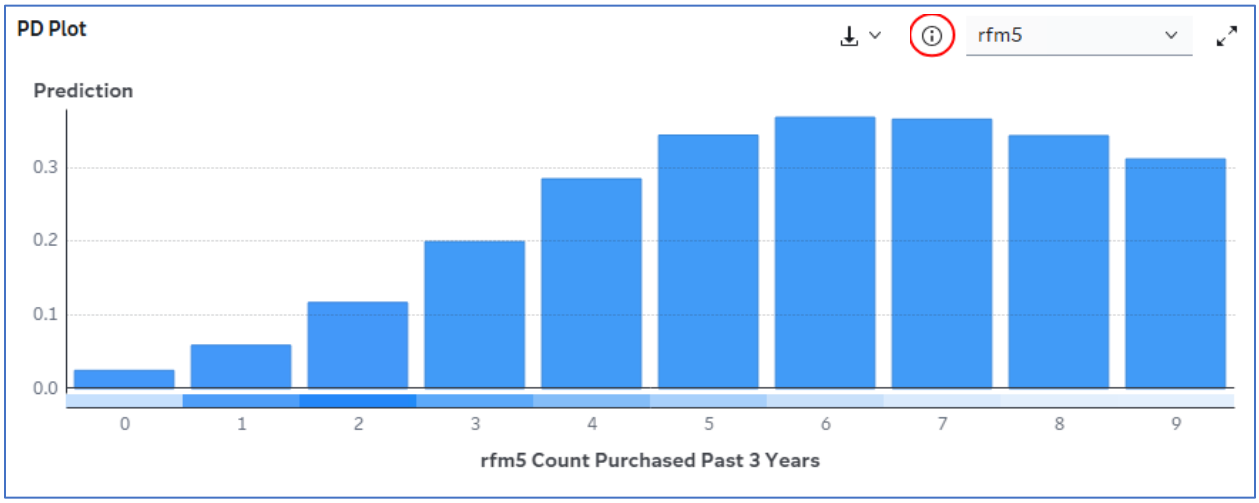
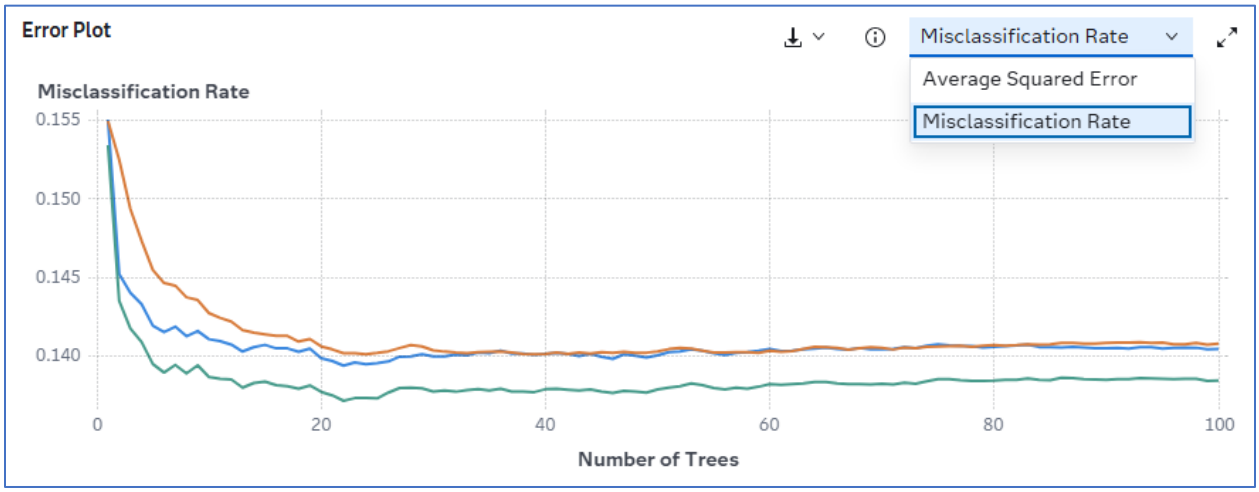


We have more demonstration topics to cover, so we don't have time to explain each of the model interpretability plots in detail. If you want to learn more about model interpretability in SAS, please refer to our tutorial video on these tools: https://www.youtube.com/watch?v=6LcyVSLwVck

22. Close the results of the Neural Network node to return to the pipeline.
23. Right-click the **Forest** node and select **Results**.
24. On the Error plot, change **Average Squared Error** to **Misclassification Rate**.

The Error plot indicates that model performance improves as decision trees are added to the forest. The menu can be used to choose between assessment statistics, misclassification rate, and average squared error.



Note: The legend on this plot shows up only if the plot is expanded.

25. Close the results of the Forest node to return to the pipeline.

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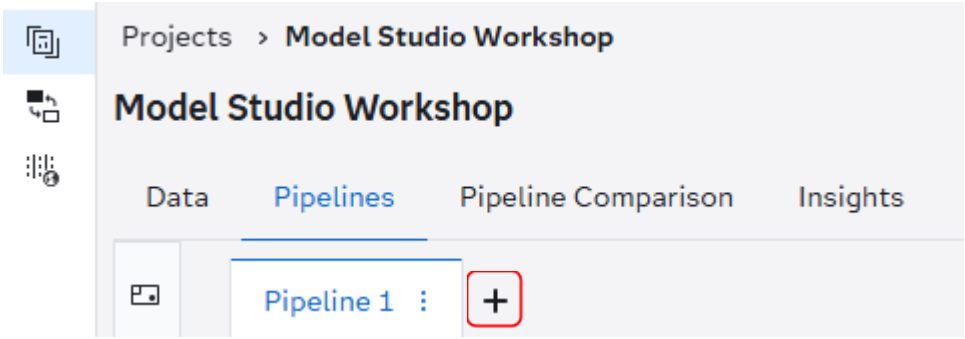
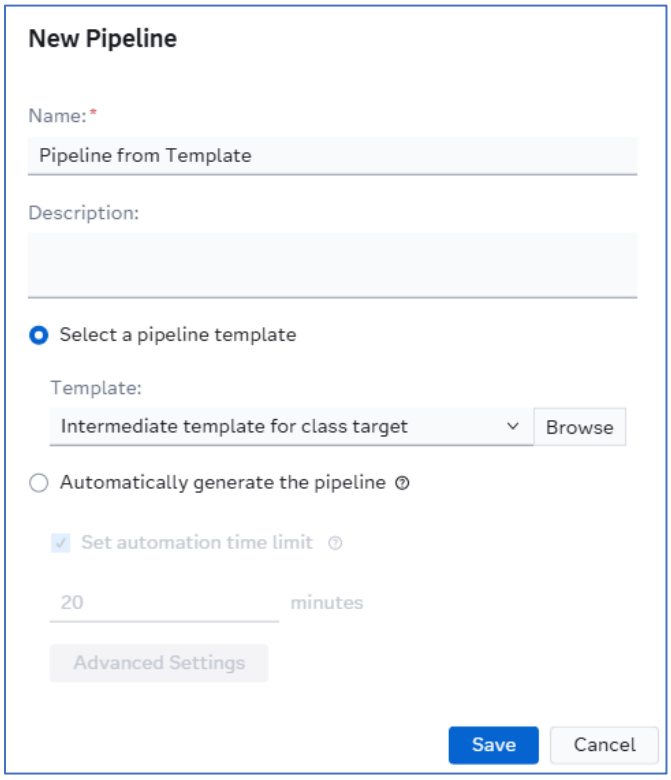
Build Models Using Pipeline Templates, Including Open-Source Models

1. Click the plus button next to **Pipeline 1** in the top left of the project to create a new pipeline.



2. Name your new pipeline **Pipeline from Template**.
3. Under the **Select a pipeline template** option, click **Browse**.
4. In the Browse Templates window, select **Intermediate Template for class target**. Click **OK**.
5. In the New Pipeline window, click **Save** to create a new pipeline from the template.



There is also the option to automatically generate the pipeline. This process uses automated machine learning to dynamically build a pipeline that is based on your data. With this option, you can

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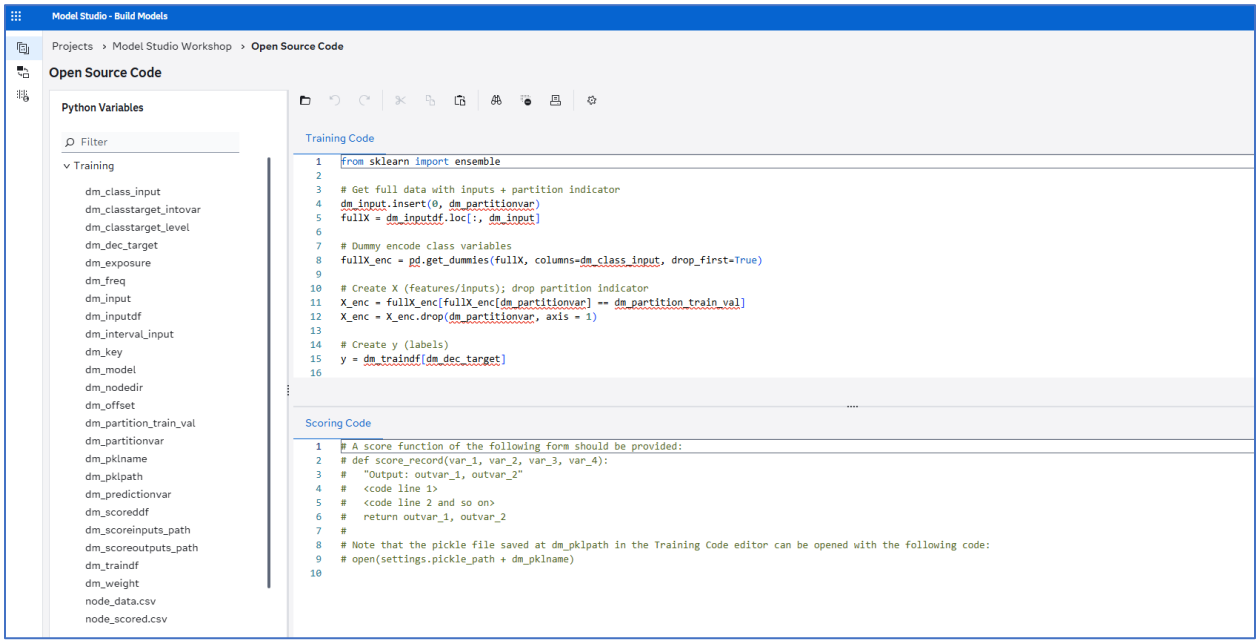
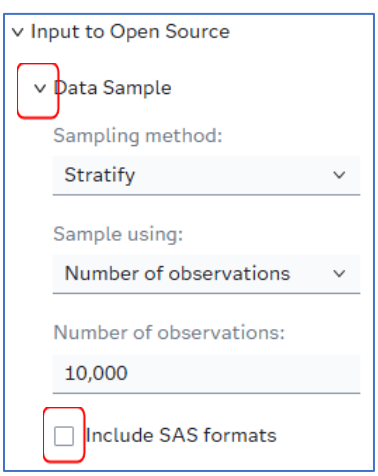
select the amount of time that the pipeline generation process is allowed to run. Be cautioned that this process can be resource intensive.

6. Right-click the **Variable Selection** node and select **Add child node > Miscellaneous > Open Source Code**.
7. In the properties pane of the Open Source Code node, select **Open code editor**.
8. Click in the Training Code window and then click the **Load Source code file** short-cut button . Navigate to **workshop (W:) > SVS**. Select **Python_Forest_Code.py** and click **Open**. Examine the Python forest code.



9. Click **Save** and then **Close**.
10. In the properties pane, expand **Data Sample** and deselect **Include SAS formats**. Also, notice that the Language property is set to **Python** (the default setting). R is another option for this property.



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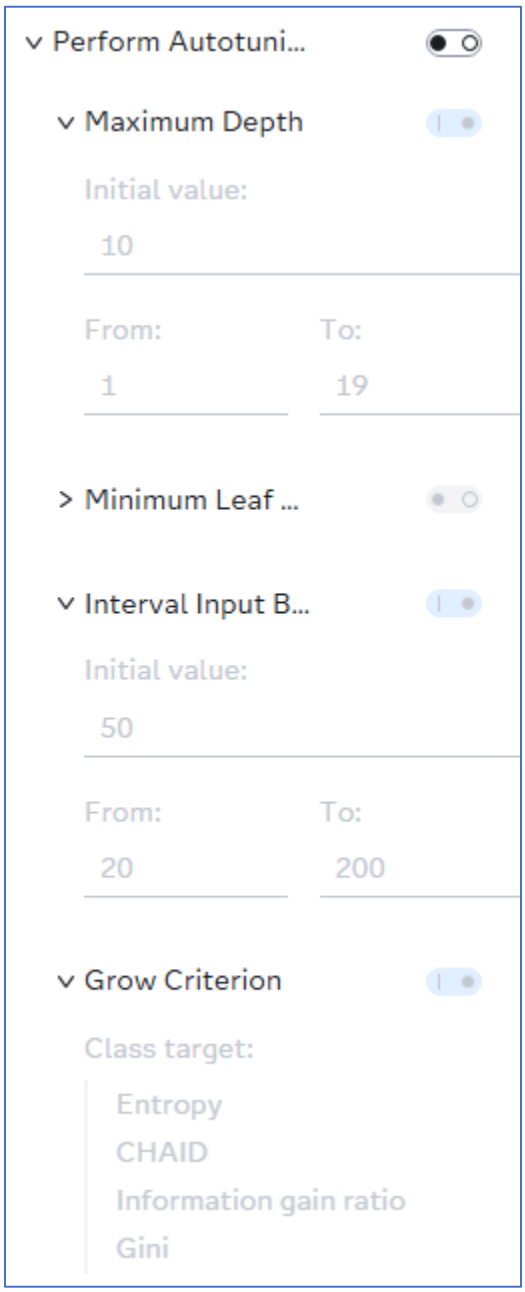
11. Right click on the Open Source Code node and select **Move > Supervised Learning.** The node changes to purple indicating it is a supervised model, and the node automatically connects to the Model Comparison node.
12. Click the **Decision Tree**. In addition to selecting options for the decision tree hyperparameters, you can take advantage of autotuning the model.



Autotuning is available for many machine learning models. For a decision tree, the tree-splitting parameters are autotuned.

11. At this time, do not perform autotuning.
12. Click **Run Pipeline** in the top right.

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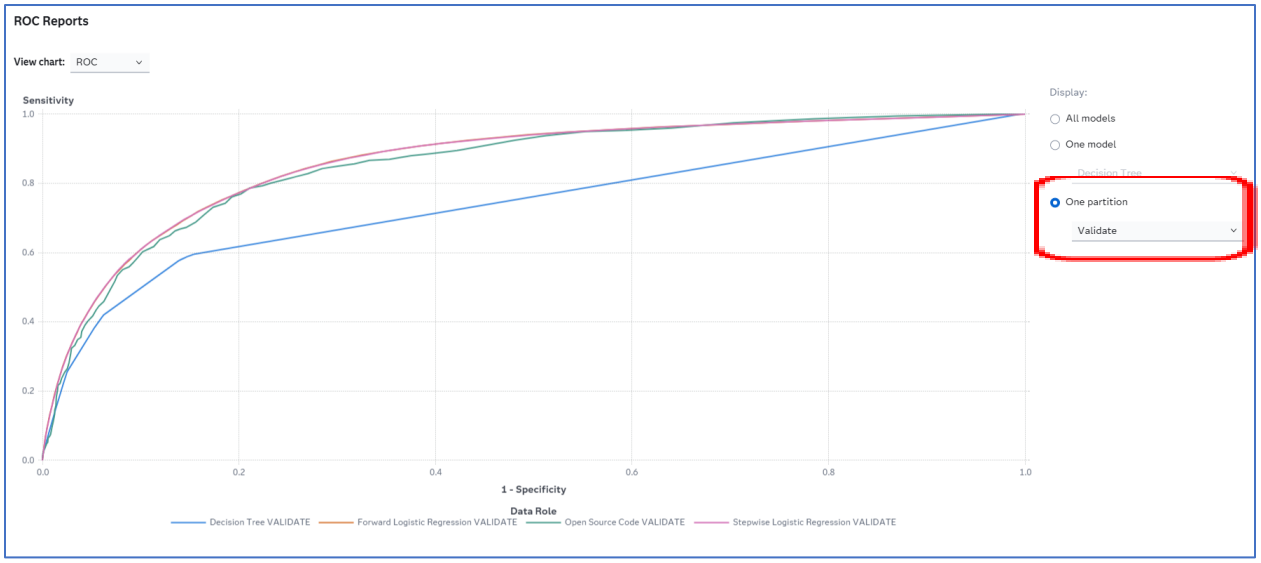
13. When the pipeline finishes running, right-click the **Model Comparison** node and select **Results**.
14. Expand the Model Comparison table to see additional fit statistics for the four listed models in this pipeline.

Model Comparison

Cham...	Name	Algorith...	KS (You...	Accuracy	Average...	Area Un...	Cumula...	Cumula...	Cutoff	Data Role	Depth	F1 Score	False Di...	False P...	Gain	Gini Coe...	ROC
*	Forward Logistic Regression	Logistic Regression	0.5779	0.8488	0.1078	0.8628	3.6328	36.3277	0.5000	VALIDATE	10	0.5307	0.3028	0.0464	2.6328	0.7256	0
	Stepwise Logistic Regression	Logistic Regression	0.5773	0.8484	0.1080	0.8626	3.6440	36.4396	0.5000	VALIDATE	10	0.5266	0.3011	0.0454	2.6440	0.7251	0
	Decision Tree	Decision Tree	0.4412	0.8345	0.1255	0.7390	3.3289	33.2895	0.5000	VALIDATE	10	0.4997	0.3701	0.0607	2.3289	0.4780	0
	Open Source Code	Open Source Code	0.5756	0.8427	0.1114	0.8541	3.5058	35.0584	0.5000	VALIDATE	10	0.5144	0.3298	0.0512	2.5058	0.7082	0

The SAS forward logistic regression model was selected as the best model based on the default KS statistic from the validation data. Notice that assessment statistics are generated and displayed for both the open source models and the SAS models.

15. Close the Model Comparison table and navigate to the Assessment tab.
16. Expand the **ROC Reports** plot. Select **One Partition > Validate**



Note: Assessment plots are created for the open source models as well as for the SAS models.

17. Close the plot and close the results of the Model Comparison node.
18. Right-click the **Open Source Code** node and select **Results**.

The Python Output window shows any output created by Python from the open-source code node. In this case, it indicates that a forest model was built and lists the settings used.

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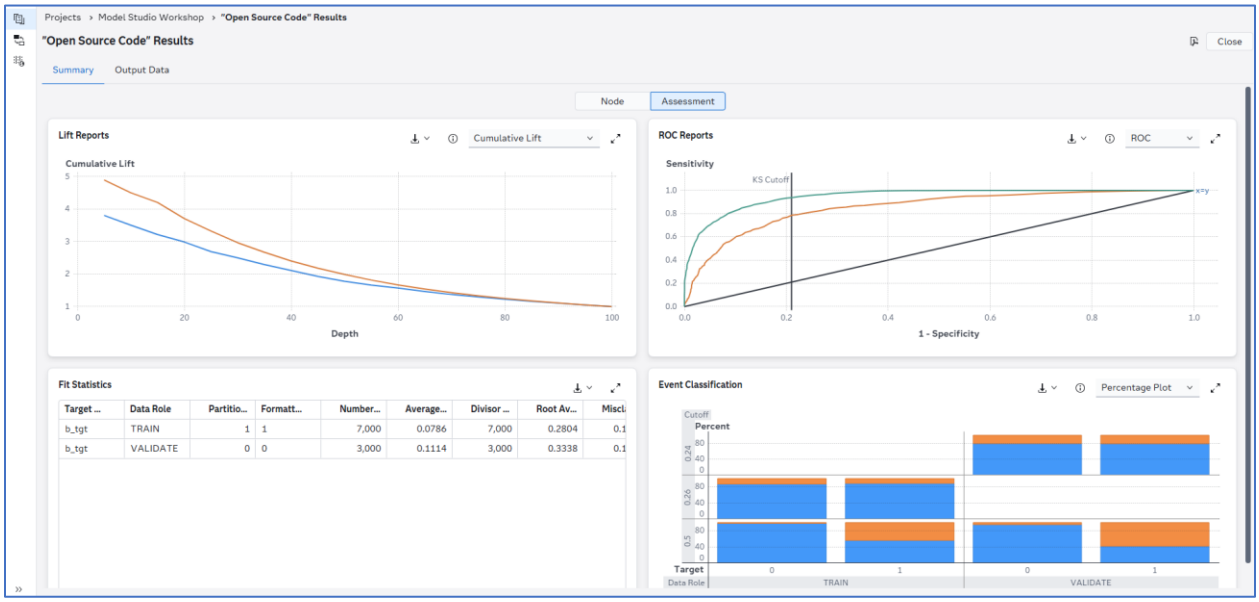
Python Output

```
RandomForestClassifier(max_depth=20, min_samples_leaf=5)
```

19. Navigate to the Assessment tab to see lift plots, ROC plots, and fit statistics for the open-source model. Notice that the Open Source Code node still produces the same assessment plots as all the other modeling nodes that SAS provides.



20. Close the results of the Open Source Code node.

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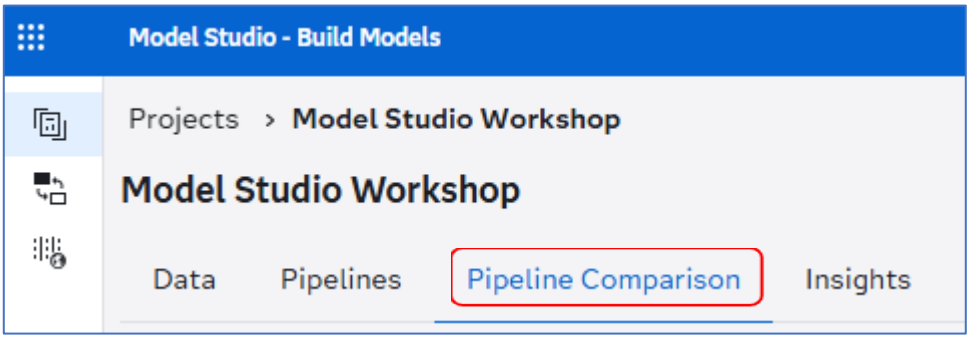
Compare Models and Assess Model Performance

1. Navigate to the Pipeline Comparison tab by selecting **Pipeline Comparison** in the top left of the project.

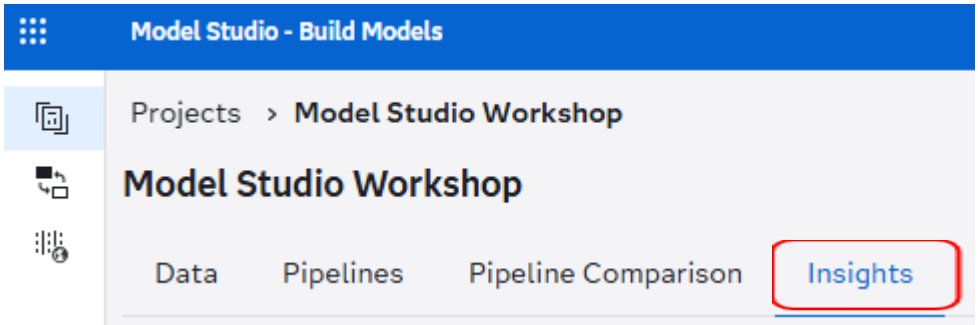


The forest model from Pipeline 1 is selected as the champion model based on the KS statistic. Results for the selected model are displayed on the Pipeline Comparison tab.

2. Select both the forest model and the stepwise logistic regression model and click **Compare**.

The same type of results that you see in the Model Comparison node appear. This time, the results compare the two selected models from the Pipeline Comparison tab.

3. Expand the **Fit Statistics** table and notice that the forest model has a validation misclassification of 14.04% and the logistic regression model has a validation misclassification of 15.12%.
4. Close the Fit Statistics table and close the results of the comparison.
5. Navigate to the project Insights tab by selecting **Insights** in the top left of the project. The Insights tab gives a summary of work done in the Model Studio project, with a focus on the champion model selected by the Pipeline Comparison tab.



The Project Summary window contains automatically generated text that explains the champion model and the most important variables used in that model.

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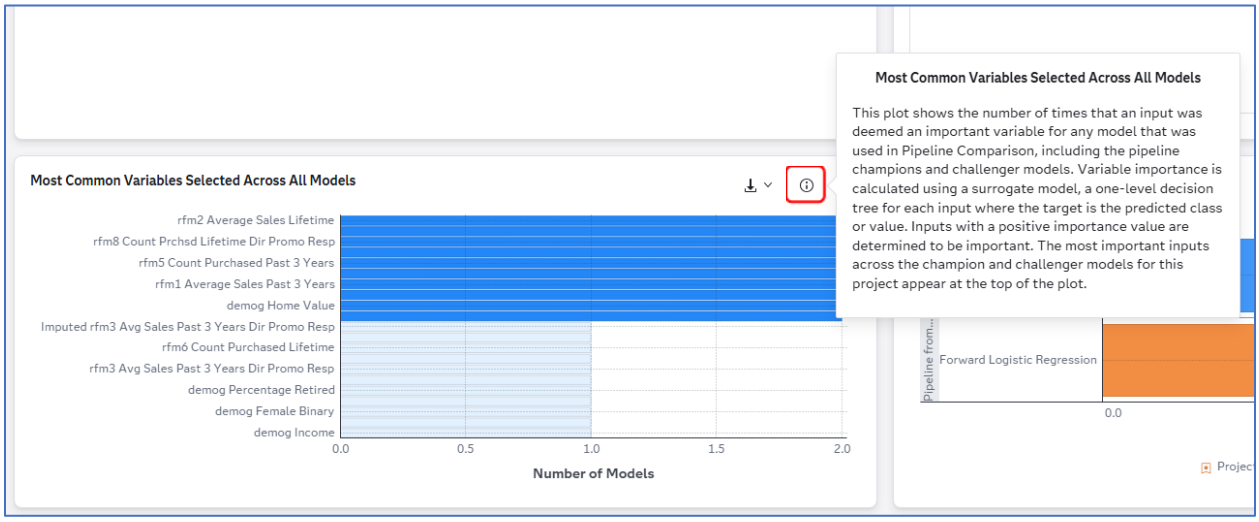
Project Summary

The champion model for this project is Forest from the "Pipeline 1" pipeline. The model was chosen based on the KS (Youden) for the Validate partition (0.6). 85.96% of the Validate partition was correctly classified using the Forest model. The five most important factors are rfm3 Avg Sales Past 3 Years Dir Promo Resp, demog Home Value, rfm1 Average Sales Past 3 Years, rfm2 Average Sales Lifetime, and rfm4 Last Product Purchase Amount.

Project Target:	tgt Binary New Product	**Project Champion:**	Forest
Event Percentage:	19.9530%	**Created By:**	Student
Pipelines:	2	**Modified:**	March 31, 2026, 09:58:51 PM

The Most Common Variables Selected Across All Models plot shows which variables are used in the models on the Pipeline Comparison tab. Note that the only models that are discussed or described on the project Insights tab are the models that appear on the Pipeline Comparison tab. Click the **View an automated description of the results** button to see an automatically generated explanation of the plot contents.



The Project Notes window enables you to write your own description or analysis of the project or its results. This is a good place to put any notes that you found about the data or the modeling effort so that people who open your project later can quickly review what you learned.

Note: In general, the project Insights tab is a good place to start when looking at a project that someone else has created. It provides a high-level overview of the best models and highlights important variables used to create those models. The Project Summary also gives you information about when the project was created and who created it.

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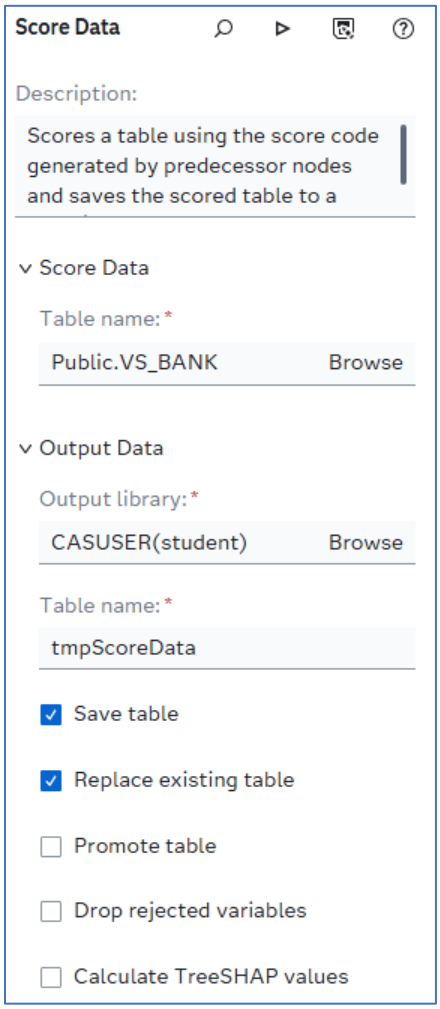
Scoring Data in a Pipeline

1. Navigate back to the Pipelines tab and select **Pipeline 1**.
2. Right-click the **Forest** node and select **Add child node** > **Miscellaneous** > **Score Data**.
3. In the options pane on the right, select **Browse** for **Table name**. Under the Available tab in the Choose Data window, select **VS_BANK**. Click **OK**. We will score the original data table with the champion forest model.
4. Select **Browse** under **Output library**. Select **cas-shared-default** > **CASUSER(student)** and then click **OK**.
5. Click the **Replace existing table** check box. Your options pane should resemble the following:



6. Click **Run Pipeline** in the top right.
7. Right-click the **Score Data** node and select **Results**. Click the **Output Data** tab.

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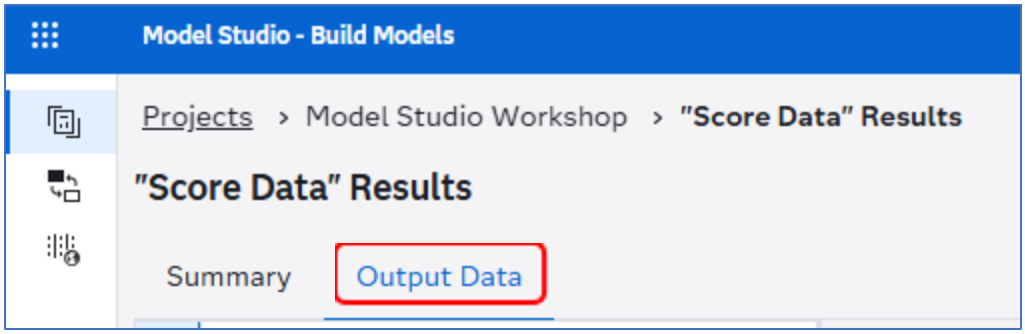
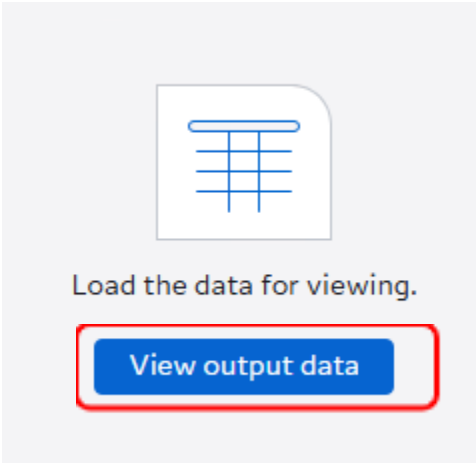


8. Select **View output data**.



9. Select **View output data** again in the Sample Data window. Do not enable sampling.
10. The **Probabilitiy for b_tgt=1** contained the scored columns of data.

Columns: 61 Rows: 1,060,038

#	Probability for b_t...	demog Customer ...	Predicted for b_tgt	demog Home Value	Probability of Clas...	demog Income	demog Percentage...
1	0.1368410054	.	0	$0	0.8631589946	$0	0
2	0.0386114545	38	0	$19,649	0.9613885455	$29,081	20
3	0.037800171	70	0	$50,774	0.962199829	$35,103	49
4	0.5883769926	17	1	$115,013	0.5883769926	$16,759	1
5	0.5331328163	49	1	$70,720	0.5331328163	$33,013	27
6	0.3996673887	.	0	$50,789	0.6003326113	$0	37
7	0.6913507921	.	1	$289,976	0.6913507921	$0	0
8	0.3518197226	63	0	$216,682	0.6481802774	$23,426	9
9	0.723116555	71	1	$137,168	0.723116555	$25,014	40
10	0.3038236371	.	0	$258,703	0.6961763629	$0	28
11	0.6731251863	.	1	$93,406	0.6731251863	$0	10
12	0.0421315382	73	0	$35,803	0.9578684618	$27,189	30
13	0.0377138488	41	0	$33,621	0.9622861512	$42,928	24
14	0.5374266853	78	1	$45,019	0.5374266853	$44,413	62

11. Click **Close** to close the Results window.

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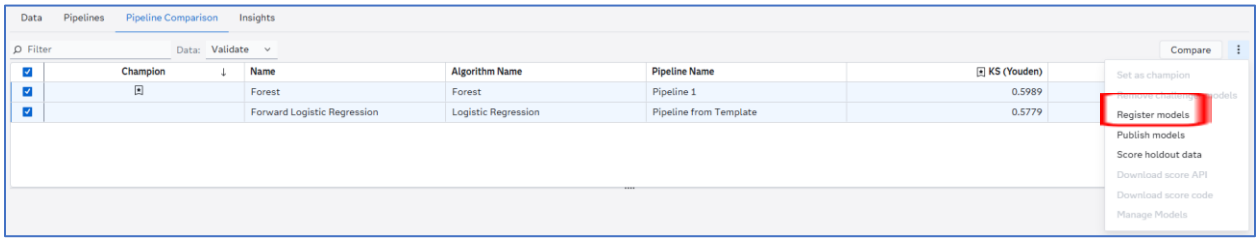
Manage and Deploy Models

1. Navigate back to the Pipeline Comparison tab.
2. Select the **Forest** model and the **Forward Logistic Regression** model using the check boxes on the left (if not already selected). Open the **Project pipeline menu** on the right and select **Register models**.

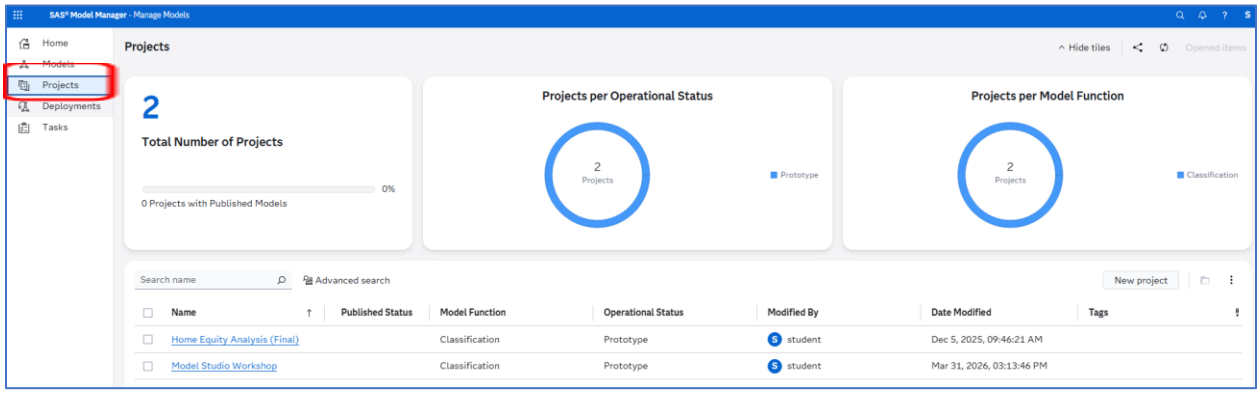


3. Click **OK** to use to the default location, **/Model Repositories/DMRepository**, to store the registered models.
4. When the activity is completed in the Register Models window and you see that the two models were registered successfully, click **Close** to close the Register Models window.
5. Click the **applications menu** icon and select **Manage Models** to open the SAS Model Manager web application.

SAS Model Manager streamlines the managing, deploying, monitoring, and operating aspects of using analytical models. It enables you to maintain a repository of SAS and open-source models for analytical projects. The repository enables easy comparison of registered models, and it has tools to monitor the performance of production models. In the event of model degradation, models can be efficiently retrained and redeployed.

With SAS Model Manager, we can automate and streamline the analytics life cycle so that models get into production faster. Automated monitoring and governance of models followed by triggering retrain, rebuild, and replace cycles ensure ongoing value to your business.

6. Select the **Projects** page, which shows that you have registered models from the Model Studio Workshop project into Model Manager.



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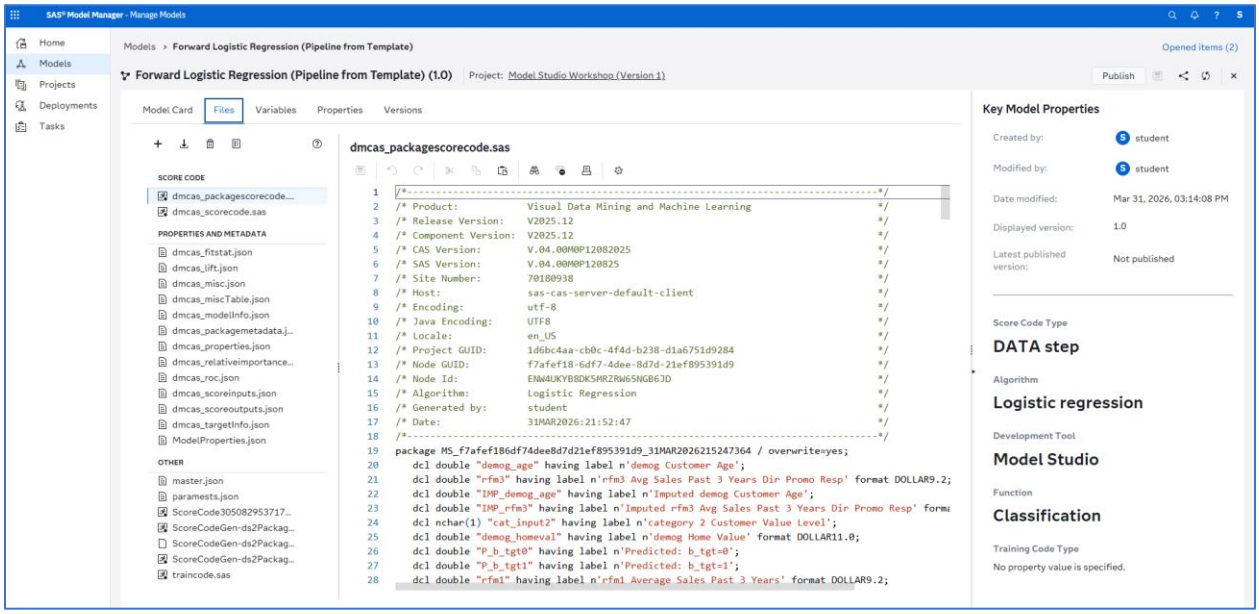
7. Select the **Model Studio Workshop** project to open it.

You have registered the forward logistic regression model and the forest model into Model Manager from your Model Studio project.

8. Select the **Forward Logistic Regression (Pipeline from Template)** model. In this demo, the logistic regression model is used, but the forest model can be chosen as well.

The **Model Card** is displayed by default.

If you click **Files,** you can explore the list of files stored for the model. The file **dmcas_epscorecode.sas** is the Base SAS score code (written in DS2) that can be used to deploy this model anywhere that SAS code can be executed. The file **dmcas_packagescorecode.sas** is score code packaged slightly differently and can be used for SAS Micro Analytic Service.



For more information about SAS Model Manager, please see [SAS Model Manager: User's Guide](#).

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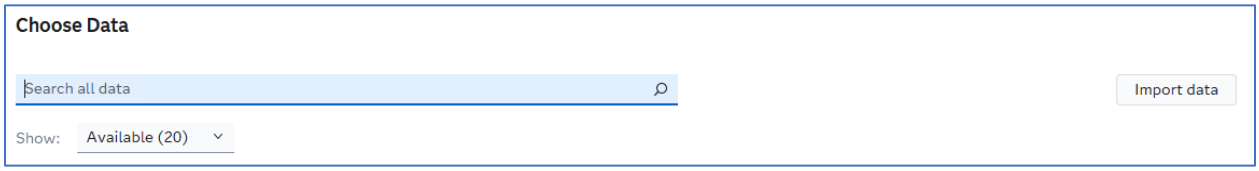
Extra Information

Loading Data into Memory

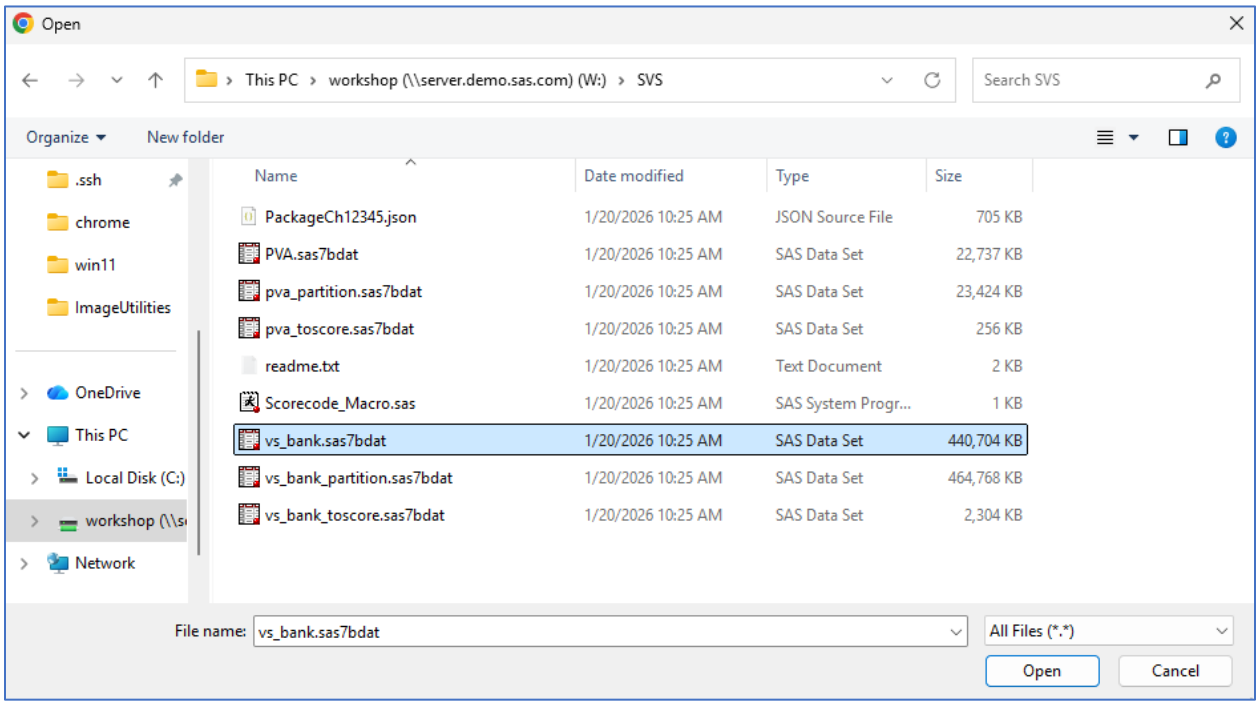
From the Choose Data window, select **Import data** to load a saved file into memory on the SAS Viya server.



1. Click on **Local files** to select a local file for import

2. Navigate to the **workshop (W:)** folder and then to the **SVS** folder.
   If you are having trouble finding the **workshop** folder you can copy the path: **W:\SVS** into the windows address bar.

3. Select the **vs_bank.sas7bdat** file and click **Open**.



4. Click **Import item** in the top right of the screen, we are loading the file **vs_bank** into the default **Public** CASLIB.

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